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Investigation of Organic Rankine Cycle (ORC) technologies in Turkey from the technical and economic point of view

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ABSTRACT

In this study, technical and economic analysis of Organic Rankine Cycle (ORC) systems in Turkey and their application areas have been examined in detail. Also, the government incentives for renewable energy productions and industrial waste heat recovery opportunities are investigated and presented. ORC technologies have been dealt with individually in geothermal, solar, biomass and industrial waste heat based power generation plants. In this concept, some power plants have been selected for each application and the technical specifications of these examinations have been presented. An application in a biomass based plant, which has 1 MW of installed capacity, has been given as an example. According to the feasibility analysis, the investment payback period of this ORC application has been calculated as 2.7 years and the results have been presented and commented.

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1. Introduction

Today, the world's primary energy demand is around 12.7 billion tons of oil equivalents (toe). According to International Energy Agency (IEA) estimations this amount will reach the values,

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Nomenclature

BEPA	biomass energy potential atlas
CHP	combined heat and power
DSI	interior ministry and the state waterworks authority
EIE	general directorate of electrical power resources survey and development administration
EMRA	energy market regulatory authority
ICE	internal combustion engine
IEA	international energy agency
kWh	kilowatt-hour
MDF	medium density fiberboard
MENR	ministry of energy and natural resources
MTA	general directorate of mineral research and exploration
ORC	organic Rankine cycle
PV	photovoltaic
TBMM	Turkish grand national assembly
toe:	tons of oil equivalents
W_e	electrical power output
WEC	world energy council
WHR	waste heat recovery
YEGM	general directorate of renewable energy

between 14.8 billion toe and 18.7 billion toe by the year 2035, with a raise of 16–47%, approximately [1]. The world's energy consumption expected to increase by more than 40% by the year 2035. Although their shares in energy demand relatively lessen, fossil fuels (oil, natural gas, coal) will continue to be the commanding resources in the next two and a half-decades with a percentage of 80%. The main reasons for growing energy needs are the industrial and urban developments, increasing life standards and the population growth, which is estimated to increase 25% in the next 20 years, especially in the countries with thriving economies. The industrial sector continues to have the biggest share of delivered energy consumption and is presumed to consume more than half of global delivered energy in 2040 [2,3]. Nuclear power and renewable energy are the world's fastest growing energy sources at the present time; each of them is increasing 2.5% per year.

Considering current policies and regulations constraining fossil fuel usage, global energy-related carbon dioxide emissions are projected to increase to 36 billion metric tons in 2020, 45 billion metric tons in 2040, a 46% increase from 2010, which was 31 billion metric tons. Economic developments are fueled by a continued dependence on fossil fuels. This accounts for greater part of the emissions raises in our planet.

Turkey has an important place in the energy markets. The country is not only a regional energy transit center but also is a developing energy producer and a consumer. Turkey's energy demand has increased rapidly over the last 15 years and most probably will continue to increase in the near future, at this rate. The crude oil consumption has also increased in the country. Turkey has very limited domestic oil reserves. So, the country imports almost all of its oil supplies. Nevertheless, Turkey plays an increasingly important role in the transit of oil because the country is strategically located at the crossroads between The Former Soviet Union Countries, the Middle East, and Europe. In addition, Turkey has a strategic role in natural gas between the world's second largest natural gas market, Europe, and the largest gas reserves of the Caspian Basin and the Middle East. Turkey's existing economic relations with its surrounding countries also verified Turkey as the most protected transit country alternative to link the regional gas market players in energy security string [4].

By restructuring of the electricity sector, both consumption and generation of electricity have expanded significantly in Turkey. Most of the electricity is generated with conventional thermal sources. Considering total primary energy consumption in the world, Turkey was ranked as the 22nd according to a report that published by Ministry of Energy and Natural Resources (MENR) of Turkey in 2012. As for Turkey's place in the energy sectors on a global scale, the country was ranked as the 20th in the world's electricity production list [5].

The ORC technology is a promising and well-proven technology, which is considered to be an important solution for low and medium temperatures (especially between 300–450 °C) heat recoveries for power production. In short, the Organic Rankine Cycle (ORC) is a heat conversion technology to convert low temperature heat into electricity. Compared to other alternative technologies, the ORC processes ensure high efficiencies for small-scale applications and/or low temperature heat sources. One of the most important advantages of ORC systems is that they can be effectively used in many areas of biomass combustion, solar desalination systems, geothermal systems and it is significantly successful when the exhaust gases of the gas turbine are at low temperatures. In addition, they are reliable, safe and flexible systems, requiring considerably low costs and maintenance requirements [6,7].

ORC systems use organic working fluids having properties like a lower boiling point, higher vapor pressure, higher molecular mass, and higher mass flow compared to water. Together, these properties enable higher turbine efficiencies than in a steam Rankine cycle [8].

There are many academic studies on ORC technologies have been published in the scientific literature. For example, Algieri and Morrone [6] aimed to analyze the energetic performances of Organic Rankine Cycles (ORCs) for small-scale applications. To that end, a parametric energy analysis was carried out to describe the proper system arrangements for a biomass power plant. In the study the results emphasize the big influence of the maximum temperature and the remarkable impact of the internal regeneration on the power plant performances. Quoilin et al. [9] presented techno-economic survey of various ORC applications. In the study, a market review was performed; technological inhibitions and optimization methods were defined and examined. The current trends in research and development for the next generation of Organic Rankine Cycles were presented extensively. Zhu et al. [10] made a theoretical research on the thermodynamic processes of a bottoming Rankine cycle for engine waste heat recovery was performed from the viewpoints of energy and exergy balance. A theoretical formula and an exergy distribution map for qualitative analyses of the essential operating parameters were presented under simplified conditions when exhaust gas was preferred as the only heat source. The results demonstrated that working fluid characteristics, evaporating pressure and superheating temperature were the essential factors influencing the system design and performances. Wei et al. [11] studied performance analysis and optimization of ORC for waste heat recovery. In the academic study, the results demonstrated that maximizing the usage of exhaust heat as much as possible was a good way to develop system output net power and efficiency; the degree of sub-cooling at the condenser outlet offered to be small (0.5–0.6 K); when the ambient temperature was too high, the system output net power and efficiency would worsen with the departure from nominal state possibly exceeding 30%. Aghahosseini and Dincer [12]

conducted a comprehensive thermodynamic analysis of the low-grade heat source Organic Rankine Cycle (ORC) and the cycle performance was examined and compared for different pure and zeotropic-mixture working fluids. Results of this analysis provide important insight into selection of the most suitable working fluids for power generating application at different operating conditions with a minimal environmental impact. Kaska [13] carried out a study on energy and exergy analysis of an Organic Rankine Cycle for power generation from waste heat recovery in steel industry. By using actual plant data, performance of the cycle and pinpoint sites of primary exergy destruction were examined in the research. Moreover, variations of energy and exergy efficiencies of the system with evaporator/condenser pressures, superheating and subcooling were presented. Guo et al. [14] performed thermodynamic analysis of a waste heat power generation system. They offered two approaches to analyze and improve ORC performance; first was to improve availability of heat source and the second was to improve the heat-work conversion of the system. Karellas and Braimakis [15] carried out a study on energy–exergy analysis and economic investigation of a cogeneration and trigeneration ORC–VCC hybrid system utilizing biomass fuel and solar power. The aim of the study was the thermodynamic modeling and economic analysis of a micro-scale tri/co-generation system capable of combined heat and power production and refrigeration, found on the joint operation of an Organic Rankine Cycle (ORC) and a Vapor Compression Cycle (VCC). Lee et al. [16] studied design and optimization of multi-component Organic Rankine Cycle using liquefied natural gas cryogenic exergy. In the study, an ORC (organic Rankine cycle) utilizing both low-grade steam from a pulverized coal power plant and LNG (liquefied natural gas) cold exergy was offered and optimized. The offered ORC was comprised of a pump, a pre-heater, an evaporator, a superheater, a reheater, a two stage turbine, and a condenser which using LNG as cold sink of the working fluid. As a result of the study, it was seen that power generation from the ternary mixture ORC was increased by about 56% and 59% as compared with the pure and binary mixture ORCs, respectively.

There are many researches on the characteristics of varied working fluids in ORC applications have been carried out in the recent years [12,17–33]. Victor et al. [20] studied on an optimization model for the composition of mixed working fluids for (Organic Rankine Cycles) ORCs and Kalina cycles. The temperatures investigated were 100–250 °C for the heat source and 30 °C for the heat sink. The optimization method of the composition was carried out with simulated annealing technique with the objective function of maximizing the thermal efficiency of the cycle, found on 1 MW of heat source. Kang et al. [21] were examined zeotropic mixtures by theoretical analysis and detailed numerical simulation in order to choose appropriate working fluids. The effects of 10 groups of mixtures on the performance of ORC were analyzed. The results showed that R245fa/R600a (0.9/0.1) was the most excellent mixture among the working fluids in the research. Le et al. [23] examined the system efficiency optimization scenarios of basic and regenerative supercritical ORCs (organic Rankine cycles) by using low-GWP (global warming potential) organic compounds as working fluid. A more familiar refrigerant, i.e. R134a, was also used to make the analogy. Thermodynamic performance comparison of supercritical ORCs using different working fluids was obtained by ranking method and exergy analysis method. Long et al. [25] studied exergy analysis and working fluid selection of Organic Rankine Cycle for low grade waste heat recovery. The optimization results indicated that the selection of working fluids depended greatly on optimal evaporation temperature. Working fluids with lower critical temperature led to a higher optimal evaporation temperature, which resulted in higher overall exergy efficiency. Wang et al. [30] suggested a double Organic Rankine Cycle for intermittent waste heat recovery. The study included calculation of optimal operation conditions of some working fluids. Pinch point analysis was also performed to analyze the ORC performance. The results of the study demonstrated that dry and isentropic fluids showed good performance. Heberle and Brüggemann [31] studied exergy based fluid selection for a geothermal Organic Rankine Cycle for combined heat and power generation. In the study, the option of combined heat and power generation was taken into account for geothermal resources at a temperature level below 450 K. Series and parallel circuits of an Organic Rankine Cycle (ORC) and an additional heat generation was investigated by using second law analysis. According to the research the most efficient concept was a series circuit with an organic working fluid that demonstrated high critical temperatures like isopentane. For parallel circuits and for power generation, fluids like R227ea with low critical temperatures were to be selected. Fernández et al. [32] studied thermodynamic analysis of high-temperature regenerative organic Rankine cycles using siloxanes as working fluids. In the study, a current of combustion gases cooled to close to their dew point temperature was taken as the reference heat source. The outlet temperature of the heat source was changed over a wide range, designating suitable fluids and schemes for each thermal level. Simple linear (MM, MDM) siloxanes in saturated regenerative schemes demonstrated good efficiencies and assured thermal stability of the working fluid. Cataldo et al. [33] proposed an approach for the fluid selection of an Organic Rankine Cycle for low-temperature waste heat recovery. The selection of an optimal working fluid was performed by an optimization process, using the genetic algorithm.

When the ORC studies examined in the scientific literature there are no related studies have been found about the potential of ORC systems and ORC applications in Turkey. In this regard, this study aims to make contribution to the literature on this topic. Therefore, Organic Rankine Cycle (ORC) technologies, ORC applications and their general view in Turkey have been investigated in detail in this academic study. Also, with the usage of ORC technologies; geothermal, solar, biomass and industrial waste heat power generation and their installed capacities have also been investigated. The results are submitted and evaluated.

2. Definition and applications of the Organic Rankine Cycle (ORC)

The Organic Rankine Cycle (ORC) is a form of energy production, essentially in geothermal, solar, biomass and waste heat recovery applications. One of the most important features of the Organic Rankine Cycle (ORC) is the possibility of using different low temperature heat sources for small and medium power generation. Because the low grade heat from these sources cannot be easily converted into electrical power efficiently by using the traditional power generation designs. In today's world, decreasing resources of fossil fuels and environmental concerns due to the climate change are the tough reasons to support clean and reliable energy production. In this direction, Organic Rankine Cycle is efficient, clean and dependable way of generating electricity.

Heat recovery is one of the most important methods to reduce fossil fuels consumption. It improves the energy sustainability and increases the energy efficiency of systems. The obtainment of heat recovery is realized while the rejected heat is available at high temperatures; electricity can be generated from the thermal power by means of solutions, like steam Rankine cycles or Stirling engines. Yet, these technologies are incompatible when the hot source temperature is below 500 °C, due to both low efficiency and technical impracticability [34]. To increase the efficiency for low temperature applications, Organic Rankine Cycle (ORC) can be

used as an alternative to the Rankine cycle. This is similar to a typical Rankine cycle except for the working fluid that drives the turbine. A high molecular mass organic which has lower boiling temperature is used as working fluid, instead of water. This makes possible to use low temperature heat sources efficiently (70–300 °C) to generate electricity in a wide range of power outputs [35]. The use of an organic fluid enacts the main difference between ORCs and traditional Rankine cycles and the choice of the working fluid is considered as a fundamental key for the maximization of the ORC global efficiency [27,28]. When selecting the fluid for ORC, the characteristics of fluid that must be taken into consideration can be listed as follows [9]:

- Thermodynamic performance,
- Isentropic saturation vapor curve,
- Freezing, boiling and stability temperature,
- Heat of vaporization and density,
- Environmental impact and safety,
- Acceptable evaporating pressure,
- Positive condensing gauge pressure,
- High vapor density,
- Low viscosity,
- High temperature stability,
- Low ozone depleting potential,
- Low greenhouse warming potential,
- Good availability and low cost.

As it is well known that geothermal energy [21,31,36–39], solar radiation [15,28,40], biomass [6,15,35,41–45] and waste heat [8,11,13,14,25,29,30,33,46–52] from industrial processes are typical energy sources for ORC low temperature practices, the Organic Rankine Cycles can also be team with internal combustion engines (ICE) or gas turbines to regain their wasted heat at medium temperature [53,54].

3. The applications and usage of geothermal, solar, biomass energies and waste heat recovery in Turkey

3.1. Geothermal energy in Turkey

Turkey has very rich geothermal energy resources and the country is ranking fifth in the world after China, Japan, USA and Iceland in geothermal heat and thermal spring applications. Turkey is geographically located in the Alpine-Himalayan belt. The geothermal potential of the country is estimated as 31,500 MW. This potential is mainly takes place in Western Anatolia around the value of 77.9%. In today's numbers, 13% of geothermal energy potential which means as 4000 MW has been made available by General Directorate of Mineral Research and Exploration (MTA) under the management of Ministry of Energy and Natural Resources. In Turkey 55% of the total geothermal areas are suitable for heating applications [55].

Since 2005, the rate of geothermal site exploration has been increased by the General Directorate of Mineral Research and Exploration. The number of discovered geothermal fields increased from 170 in 2005 to 187 in 2008 [56]. As it can be seen in Table 1, geothermal energy usage in Turkey has reached the amount of 114.2 MWe at the end of 2012 [55].

In addition to geothermal resources, there are 1000 hot water and mineral water sources and wells available in Turkey. 170 of these wells have the temperature above 40 °C. 13 of these sources are considered the most valuable ones by the sector for the generation of electricity. The potentials of geothermal electricity generation and source temperatures of these regions are given in Table 2 [55].

Table 1
Geothermal energy use in Turkey (at the end of 2012) [55].

Usage	Capacity (MW)
Electricity generation	114.2 MWe
Residence heating	805 MWt
Thermal tourism	402 MWt
Greenhouse heating	507 MWt

Table 2
The potentials of geothermal electricity generation and the source temperatures of geothermal regions in Turkey [55].

Geothermal region	Temperature (°C)	Potential of electric energy generation (MW)
Denizli Kizildere	200–242	80
Aydin-Germencik	200–232	130
Manisa-Alaşehir-Kavaklidere	213	15
Manisa-Salihli-Gobekli	182	15
Canakkale-Tuzla	174	80
Aydin-Salavatli	171	65
Kütahya-Simav	162	35
İzmir-Seferihisar	153	35
Manisa-Salihli-Caferbey	150	20
Aydin-Sultanhisar	145	20
Aydin-Yilmazkoy	142	20
İzmir-Balcova	136	5
İzmir-Dikili	130	30
Total		**550**

The installed capacity of newly commissioned power plants based on renewable energy resources has been around 2 002 MW in the first nine months of the year 2014. The share of geothermal sourced electricity generation power plants is 47.6 MW [1]. Currently, there are 14 geothermal power plants are in the energy sector in Turkey. While geothermal installed capacity was only 17.5 MW in 2002, it has reached the value of 358.4 MW as of today's numbers. Compared to 2002, the applications of geothermal greenhouse heating has reached the value of 3130 thousand square meters with an increase of 526% as of September 2014 and also the residential heating has reached number about 89,563 dwellings with an increase of 198%.

The private sector began to take part in geothermal energy production in Turkey with the Law Number 5686: Geothermal Resources and Natural Mineral Water Act came into force in 2007. Between the years 2008 and 2013, total 96 pieces of geothermal land have been transferred to investors, 16 pieces of these land are for electricity generation and 80 of them for the purpose of heating and thermal tourism. So, total tender price of this geothermal land is $ 545 million, brought to the country's treasury [1].

3.2. Solar energy in Turkey

Because of its geographical location Turkey has a higher solar energy potential than most of the other EU (European Union) countries. According to the General Directorate of Electrical Power Resources Survey and Development Administration (EIE), the annual average total insolation duration is determined as 2640 h (7.2 h/day) and the annual average solar radiation is determined as 1311 kWh/m^2-year (3.6 kWh/m^2-day) for Turkey. The solar energy potential of Turkey is calculated as 380 billion kWh/year. Turkey's Total Solar Radiation (kWh/m^{2}year) can be seen in Fig. 1 [57] and the solar energy potential of Turkey and corresponding insolation durations are given in Table 3 on a monthly basis [57].

Solar energy utilization in Turkey is mainly composed of hot water generation systems which turn solar energy in to thermal

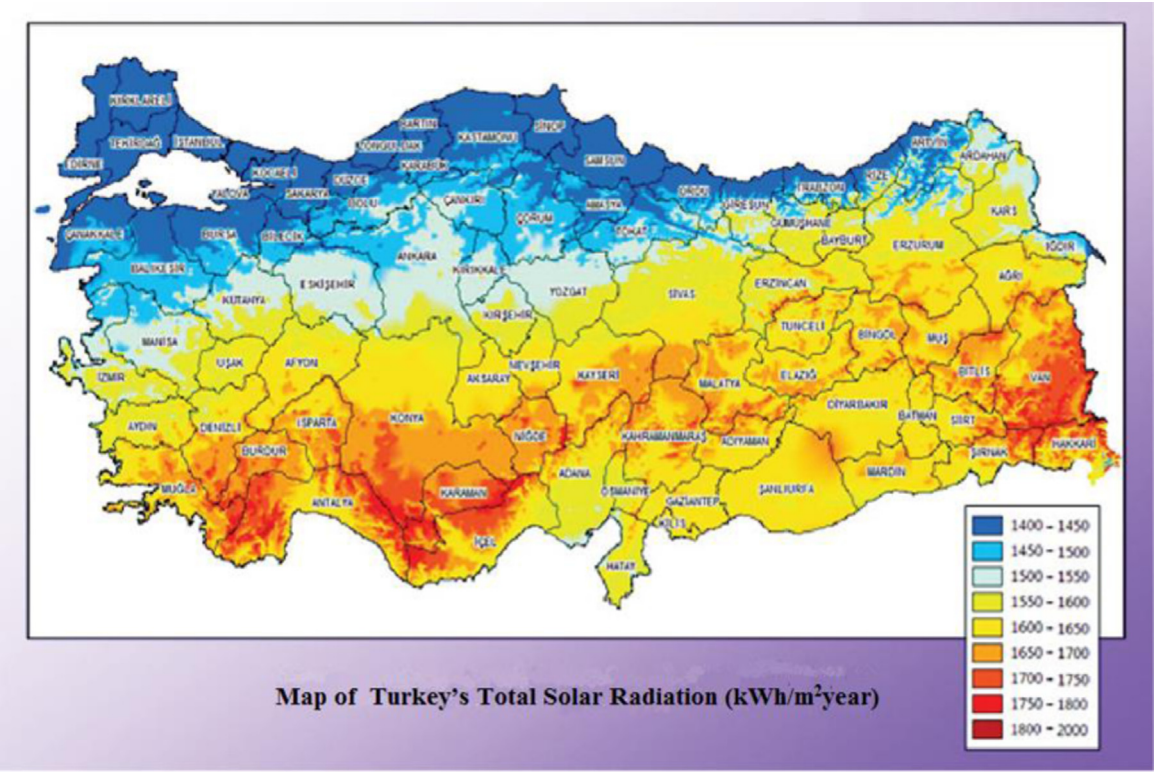


Fig. 1. Turkey's total solar radiation (kWh/m^{2}year) [57].

Table 3
Turkey's average potential of solar energy and insolation durations on a monthly basis.

Months	Total monthly solar energy[a]		Insolation duration[a] (h/month)
	kcal/cm^2-month	kWh/m^2-month	
January	4.45	51.75	103.0
February	5.44	63.27	115.0
March	8.31	96.65	165.0
April	10.51	122.23	197.0
May	13.23	153.86	273.0
June	14.51	168.75	325.0
July	15.08	175.38	365.0
August	13.62	158.40	343.0
September	10.60	123.28	280.0
October	7.73	89.90	214.0
November	5.23	60.82	157.0
December	4.03	46.87	103.0
Total	112.74	1311	2640
Average	308.0 kcal/cm^2-day	3.6 kWh/m^2-day	7.2 h/day

[a] Data source: EIE (2010) [57].

energy. Solar energy is generally used in the Mediterranean and Aegean regions in Turkey. Total installed area of solar collectors is 12,000,000 m^2 and annual production capacity is 750,000 m^2. While annual thermal energy production from solar energy was around 210,000 toe (tons of equivalent) in 1998, it reached 420,000 toe in 2007. Considering last decade's developments, Turkey is regarded as an important solar collector producer and solar energy user. In addition to solar collectors, installed capacity of solar batteries is around 1 MW, mostly used by public institutions to meet small power necessities and for research purposes (EIE, 2010) [57].

The share of solar energy utilization is planned to be increased at the maximum level, within the scope of increasing renewable energy production in energy supply in Turkey. After the completion of legal necessities and technical regulations, the Energy Market Regulatory Authority (EMRA) took 496 the license applications for the production of electricity from solar energy in June, 2013. These applications correspond to 9000 MW installed power, approximately [1].

3.3. *Biomass energy in Turkey*

Biomass is organic substance that has collected sunlight in the form of chemical energy. All biomass is produced by green plants converting sunlight into plant substance through photosynthesis [44]. Biomass can be used to meet a variety of energy needs, including generating electricity, heating homes, fuel for vehicles and supplying process heat for the industrial facilities.

The Conversion of biomass to energy comprises of two main process technologies: thermo-chemical and bio-chemical/biological. Mechanical extraction is the third technology for producing energy from biomass. Three main processes are used for the thermo-chemical conversion of biomass. These are combustion, gasification and pyrolysis. Bio-chemical conversion has two major processes; fermentation and anaerobic digestion [45].

Renewable energy resources in Turkey are the second largest source behind coal in terms of energy generation, and an important portion of the renewable energy generation is met by biomass [58]. The annual biomass potential of Turkey is around 32 Mtoe. The current and projected biomass energy production in Turkey is presented in Table 4. Almost all of the biomass energy in the country is consumed in residences mostly for cooking and heating purposes. Wood is the main heating fuel for 6.5 million residences

Table 4
Present and projected biomass energy production in Turkey [56].

Years	Total biomass production (ktoe)[a]
2000	6982
2005	7260
2010	7414
2015	7320
2020	7520
2025	7810
2030	8205

[a] Data source: [56].

Table 5
Biomass sources and their energy potentials in Turkey (WEC, 2007)* [59].

Resource	Raw material potential (million tons)
Municipal solid wastes	25
Wood	3.52
Forestry/wood processing	3.56
Agricultural residues-straw + stalk	13.2
Agricultural residues-seed, shells, wood chips	4
Fertilizers	13.8

* WEC, World Energy Council, Survey of Energy Resources, 2007 [59].

Table 6
Turkey's annual biomass energy potential [43].

Type of biomass	Energy potential (Mtoe)	Annual potential (Mt)
Annual crops	14.9	55
Perennial crops	4.1	16
Animal wastes	1.5	7
Forest residues	5.4	18
Residues from agro industry	3.0	10
Residues from food industry	1.8	6
Other	1.3	5
Total	32.0	117

in Turkey. Paper and pulp industries also use waste wood and liquor for their fuel necessity, which is around 60% [56].

Table 5 shows the biomass sources of Turkey and corresponding energy potentials (WEC, 2007) [59].

While Turkey's installed capacity was 12,305 MW in renewable energy sources in 2002; this value has reached the number of 27,585 MW at the end of September in 2014. The electricity generation from renewable sources was 34 billion kWh in 2002, this number reached to 69.5 billion kWh in 2013, with an increase of 104%. According to the recent numbers, the installed capacity of newly commissioned power plants based on renewable energy sources for landfill gas, biomass and waste heat sourced electricity generation power plants is 64.1 MW in total [1].

The studies for the preparation of "Biomass Energy Potential Atlas (BEPA)" have been continuing in order to determine the biomass potential of Turkey. This atlas has been prepared by General Directorate of Renewable Energy under the Turkish Ministry of Energy and Natural Resources.

Turkey has great biomass potential to provide rural energy services based on forest and agricultural residues. Among the renewable energy resources, biomass is an important source in Turkey because its share of total energy consumption is relatively high. Table 6 shows Turkey's annual biomass energy potential [43].

Table 7
Energy savings potentials in industrial sectors in Turkey [61].

Sectors	Energy Saving Potentials (%)		Energy Saving Potentials (10^3 TOE/year)
	Electricity	Fuel	
Iron and steel	21	19	1402
Cement	25	29	1124
Glass	10	34	261
Paper	22	21	206
Textile	57	30	1097
Food	18	32	891
Chemistry	18	64	2283
Others	–	–	729

According to data that received from EMRA (the Energy Market Regulatory Authority), in Turkey, operating capacity of power plants that generating electricity from biomass is around 138.7 MW in the business. The capacity of power plants that under construction for biomass energy generation is 13.4 MWe. Power plants capacity for biomass energy generation is 45.8 MWe, which is in the project planning phase, at present. Turkey's energy generation plants from biomass can be grouped as follows;

- Energy production from waste gas,
- Power generation from organic waste through anaerobic fermentation,
- Energy production from biomass in the combustion plants,
- Energy production from organic wastes by using the method of gasification.

3.4. Waste heat recovery potential in Turkey

When examining the energy consumption in Turkey, it can be seen that 36% of total energy consumption was realized by the industrial sectors in 2010. Main industries that consumed this energy amount extensively were iron and steel, cement, ceramics, glass, non-metallic mineral. According to 2010 data, the iron and steel sectors had the highest level of energy consumption amount with the usage of 22%. After iron and steel industries, non-metallic mineral and earth sectors such as cement, glass, ceramics, brick were the second energy consumers in the industry with a share of 19%. Another big energy consumption area was the paper industry in 2010. The paper industry's energy consumption rate was 3%. Among other major energy consumers, the chemical industry had the share of 12%, the food and textile industries had the share of 9% each, in the industry [60]. The production costs of iron and steel industries range between 20% and 50% in the total energy costs as the most intensive energy users in Turkey. In these sectors, some industry organizations have reached the level of energy intensity that can compete with the world's other companies. Also, their energy efficiency enhancement studies have been increased significantly in recent years. However, there is still huge energy savings potential and opportunities in these sectors need to be carried out [60].

The evaluation of energy saving potentials plays an important role in reducing energy costs. There are different saving potentials for different sectors to increase energy efficiency in Turkey. The energy audits that conducted in these industrial facilities have shown that the average energy saving potential is at least 20% in these sectors. The energy audits have also shown that approximately 50% of this potential does not require significant investment and their average payback period is less than two years [60]. In addition, approximately 5.7 million toe energy saving is possible in Turkey's industry in general. Energy saving potentials in industrial sectors in Turkey can be seen in Table 7 [61].

4. The current situation of Organic Rankine Cycle applications in Turkey

In Turkey, ORC applications has been implementing for the purpose of electricity generation from biomass, geothermal and waste heat for the last five years. These applications are becoming widespread due to the government's feed in tariffs which guaranteed purchase prices for renewable energies generated at different scales and through different sources for up to 10 years. Therefore, ORC applications are expected to increase in the near future.

4.1. ORC applications for biomass power generation in Turkey

In Turkey, there are two groups of sectors come to the fore for the purpose of electricity generation from biomass through ORC applications. First group of these sectors are power generation plants, which use landfill gas to produce power and the second group of plants use the method of waste incineration (wood chips, sawdust, and bark etc.) to produce power through the ORC applications. There are fourteen landfill gas combustion plants in operation in Turkey. These facilities have an installed capacity of about 100.42 MW in total, but there is only one plant that using ORC technology to generate electricity. There are also gas engines with the capacity of 25 MWe in this plant. ORC units are used for the purpose of power generation by getting heat from exhaust gases of these gas engines and cooling water. With these ORC applications 5–10% more electricity can be produced in addition to the existing electricity production in that plant. As it can be seen in this example, ORC applications are expected to increase in landfill gas combustion plants in Turkey [1].

In this study, an industrial facility that manufactures MDF (Medium Density Fiberboard) and chipboard in wood sector in Turkey has been chosen as an example for an ORC application. Waste like wood chips, sawdust, bark, etc. is combusted in a thermal oil boiler to get thermal oil. A portion of thermal oil is used for the heat requirement of the process and the rest is used in the ORC unit to produce hot water and electricity. The produced electricity is sold to the grid by means of renewable energy incentives. Hot water is used for space heating in the winter and it also meets the needs of cooling for the process through a water-cooled absorption chiller in the summer. The technical properties of the ORC unit that used for power generation in the example industrial wood sector are given in Table 8.

There are 34 biomass power plants in business in Turkey and it is possible that ORC systems can be applied on them successfully. When the types of these plants are studied; some of these plants are using landfill gas in internal combustion engines (ICEs), some of them are producing biogas through anaerobic digestion and the rest of them are biomass incineration plants. The total installed capacity of these biomass plants is 195 MW. About 25 MW of electricity can be produced by applying the ORC systems in these biomass power plants. However, in the current situation, the installed capacity of the ORC power plants is around 6.5 MWe. It is possible to make this figure three times more by applying ORC systems in the biomass based power plants. In addition, it is foreseen that the number of ORC power plants increase rapidly in the condition of continuation of incentives given by the state. When making a tender offer for a landfill gas incineration plant that planning to be built, the installation of the ORC systems in the plant should be recommended or made compulsory by the state, if its installed capacity over 1 MWe. This will also assure the efficient use of renewable energy sources and make great contribution to the country's economy.

Table 8
The technical properties of the ORC unit that used for power generation in the example industrial wood sector.

Parameters	Values
Hot oil temperature (ºC)	280–300
Hot water outlet temperature (ºC)	93
Electricity generation capacity (kWe)	985
Electricity generation efficiency (%)	20

4.2. ORC applications for geothermal power generation in Turkey

In Turkey, 60% of total geothermal resources is suitable for the use of heating, 30% is for the tourist attractions etc. and 10% is suitable for the production of electricity economically. When geothermal fields examined in Turkey, the temperature of these resources have high and middle degrees in western Anatolia. As for the central and eastern Anatolia the temperature of the geothermal fields can be measured as from the mid-to lower degrees. Also, it is well known that there are many wells that are unsuitable for power generation with large capacity in Turkey. These wells are remaining inactive or only used for heating purposes and at a temperature range from 90 to 125 °C. In fact, those resources that can be evaluated for power generation with the usage of Organic Rankine Cycle and therefore, they can be brought into the economy, successfully.

Generally, geothermal investments are seen as the investments that have "high uncertainty" until their production stage. In the case of such uncertainty, one of the healthiest investment models is called as "Modular growth". In a modular growth model, the investment can be realized in a short time in the form of small modules. ORC technology can be easily applied to a modular growth model.

In Turkey there are various double-fluid (binary) ORC applications in geothermal plants can be found with high MW capacities in Turkey. Binary power plants use geothermal resources, which the temperature of these sources ranges between 150 °C and 180 °C. These power plants have been put into practice for 25 years; still they are developing their technologies. The technical properties of the ORC binary unit in the exemplary geothermal power plant in Turkey are given in Table 9.

The installed capacity of electricity generating plants from geothermal energy through ORC applications is 197 MWe in Turkey. The installed capacity of the geothermal based ORC plant that under construction is about 98 MWe. If the incentives for electricity production from geothermal sources continue to be given to the related sectors, it is strongly believed that the use of the ORC technology in geothermal based energy productions can rapidly spread in Turkey in the immediate future.

4.3. ORC applications for waste heat power generation in Turkey

After being released from a process of a production, the waste heat that has medium and low temperature can be used to meet

Table 9
The technical properties of the ORC binary unit in the exemplary geothermal power plant in Turkey.

Parameters	
Gross electricity generation (MW)	7.36
Net electricity generation (MWe)	6.5
Reservoir temperature (ºC)	176
Heat sources	Brine and steam
Organic fluid	N-Pentan
Condenser type	Air cooled condenser
Re-injection temperature of geothermal fluid (ºC)	80–90

Table 10
The technical properties of the example ORC unit that are used for the production of electricity from waste heat in Turkey [13].

Parameters	–
Gross electricity generation Gross elektrik üretimi (kW)	262
Net electricity generation Net elektrik üretimi (kWe)	250
Evaporator pressure (bar)	10.8
Condenser pressure (bar)	2.1
Heat sources	Steam from slab reheat furnace
Organic fluid	R245fa

the need of the processes heat or space heating in the industry. In addition, with the ORC technology electricity production can be realized from the medium to low temperature waste heat sources. If waste heat is not used to meet the need of process then it is more suitable method to use for the electricity production through an ORC unit. As it is mentioned above, the production of electricity from waste heat with the ORC applications is very new for Turkey, but it is improving, gradually. The technical properties of the example ORC unit that are used for the production of electricity from waste heat in Turkey can be seen in Table 10 [13].

The installed capacity of the plants that generating electricity from industrial waste heat through the ORC applications is below 1 MWe in Turkey. The reason for the failure of spreading ORC applications in Turkey is the lack of incentives for waste heat based power generation. Due to this reason, the simple payback period of this kind of investment is around 5–8 years, which is quite long. Also, the high cost of the ORC systems productions is another obstacle. Unfortunately, unless the government's incentives for electricity generation from waste heat is assured and the decrease of the investment and production costs of the ORC systems is realized, electricity generation from waste heat through ORC applications cannot be widespread in Turkey.

5. Renewable energy production incentives in Turkey

At the end of 2010 Turkish Grand National Assembly (TBMM) passed a new renewable energy bill determining regulations and feed-in tariffs in the sector (Law no: 6094) [58]. According to the law, producers of renewable energy who started operation between May 18, 2005 and December 31, 2015 will be guaranteed power purchase prices for a period of ten years. Wind and hydroelectric power will be bought at USD 0.073 per kilowatt-hour (kWh), the price for geothermal energy is defined at USD 0.105 per kilowatt-hour (kWh), and biomass, including landfill gas and solar energy will be purchased at USD 0.133 per kilowatt-hour (kWh). However, the law limits the solar power capacity eligible for support to 600 MW until December 31, 2013 and entitles the Council of Ministers to determine the capacity after that. The Council of Ministers is also authorized to designate new purchase prices, which should not exceed the current prices, for facilities established after December 31, 2015. Renewable energy producers that got their operation licenses before December 31, 2015 are entitled to receive additional subsidy ranging from USD 0.004 to USD 0.035 per kWh for a period of five years if they use locally-produced equipment and technology for their plants [58]. License procedures will be carried out by the Energy Market Regulatory Authority (EMRA) in cooperation with The Ministry of Energy and Natural Resources, The Ministry of Interior and the State Waterworks Authority (DSI). According to this new renewable law, the incentives for the renewable energy productions are given in Table 11 [58]. Additional support amounts ("Local Content Additions") for the companies that use locally produced equipment and components are given in Table 12 [58].

Table 11
Renewable energy (Law no: 6094) incentives in Turkey [58].

The plant type of the generation of energy from renewable resources	The prices that will be applied (USD cent/kWh)
A hydroelectric power plant	7.3
A wind power plant (a wind farm)	7.3
A Geothermal energy plant	10.5
A biomass supplier (including landfill gas)	13.3
A solar energy plant	13.3

6. Feasibility analysis of the ORC applications

A feasibility study of the ORC unit can differ in factors such as the temperature of the heat source, the continuity of the heat source, the annual working time, the price of sales per unit for electricity and heat. To give the information about the feasibility study of ORC systems in Turkey, an ORC unit has been chosen as an example, which uses biomass as a heat source. The specifications of this ORC unit are given in Table 13. Also, the feasibility analysis of this ORC unit can be seen in Table 14.

7. Conclusions

The Organic Rankine Cycle (ORC) is a heat conversion technology to convert low temperature heat into electricity. ORC technology assures to recover heat from different energy sources, both at low and medium temperature. This technology allows taking advantage of essential increase in the supply of renewable energy for today's international energy production race. Also, it has many opportunities in heat recovery from different industrial energy applications to improve power productions for Turkey.

In this academic study, technical and economic analysis of Organic Rankine Cycle (ORC) systems and ORC applications in Turkey has been examined. ORC technology is used for power generation from heat sources like industrial waste heat, solar ponds, biomass combustion, geothermal heat that in medium and low temperatures. This heat is converted into useful work to generate power and heat. ORC systems use organic working fluids (mainly hydrocarbons and refrigerants), which permits to use Organic Rankine Cycle in heat recovery applications from the sources that have low temperatures.

Turkey is an energy importing country because its fossil fuel reserves are very limited. So, the country needs to use its domestic sources, such as renewables and other sources, in order to decrease dependency on other countries for the supply of fossil fuels [62].

Turkey has an important place in the energy markets. The country is not only a regional energy transit center but also is a developing energy producer and a consumer. Turkey's existing economic relations with its surrounding countries also verified Turkey as the most protected transit country alternative to link the regional gas market players in the energy security string.

In Turkey, there are ORC applications in power plants that use biomass, geothermal and at low and medium temperature waste heat as fuels to produce electricity and heat. ORC applications have started to be carried out over the last five years in Turkey. ORC applications are expected to expand in the coming years due to the incentives for electricity production from renewable energy

Table 12
Additional support amounts for the companies that use locally produced equipment and components [58].

Plant Type	Locally produced equipment and components	Domestic contribution supplement (USD cent/kWh)
A – Hydroelectric power plant	1. Turbine	1.3
	2. Generator and power electronics	1.0
B – A wind power plant (A wind farm)	1. Propeller	0.8
	2. Generator and power electronics	1.0
	3. Turbine Tower	0.6
	4. Rotor and all mechanical components in the nasel group (the exception of payments made for propeller group with generators and power electronics.)	1.3
C – Facilities of photovoltaic solar panels	1. Integration of PV panels and manufacture of the structural mechanics of the solar PV panels	0.8
	2. PV modules	1.3
	3. Cells that make up PV modules	3.5
	4. Inverter	0.6
	5 Beam materials that focusing solar energy on the solar PV module	0.5
D – The plants that based on concentrated solar power	1. Radiation pick-up tube	2.4
	2. Reflective surface plate	0.6
	3. Solar Tracking System	0.6
	4. The mechanical parts of the thermal energy storage system	1.3
	5. Mechanical parts of the steam production system by collecting the solar beam on the tower	2.4
	6. Stirling engine	1.3
	7. The panel integration and structural mechanics of the solar panel	0.6
E – Facilities that based on biomass energy	1. Fluidized-bed steam boiler	0.8
	2. Liquid or gas-fired steam boiler	0.4
	3. Gasification and gas cleaning group	0.6
	4. Steam or gas turbine	2.0
	5. Internal combustion engine, or Stirling engine	0.9
	6. Generator and power electronics	0.5
	7. Cogeneration system	0.4
F – Facilities that based on geothermal energy	1. Steam or gas turbine	1.3
	2. Generator and power electronics	0.7
	3. Steam injector or a vacuum compressor	0.7

Table 13
The technical properties of the ORC unit that uses biomass as a heat source.

Parameters	–
Net electricity generation (kWe)	995
Mechanical power (kW)	1034
Biomass thermal power (kW)	6476
Thermal power to the condenser (kW)	5440
Biomass (bark) energy content (kW/kg)	2325
Annual operating hours (h/year)	8000
Thermal oil boiler efficiency (%)	85

sources. In Turkey, electricity production from renewable energy sources is subsidized by the Turkish government.

There are two group of sectors come to the fore for the purpose of electricity generation from biomass through ORC applications in Turkey. First group of these sectors are power generation plants, which use landfill gas to produce power and the second group of plants use the method of waste incineration (wood chips, sawdust, and bark etc.) to produce power through the ORC applications. There are fourteen landfill gas combustion plants in operation in Turkey. These facilities have an installed capacity of about 100.42 MW in total, but there is only one plant that using ORC technology to produce power. There are also gas engines with the capacity of 25 MWe in this plant. ORC units are used for the purpose of power generation by getting heat from exhaust gases of these gas engines and cooling water. With these ORC applications, 5–10% more electricity can be produced in addition to the existing electricity production in that plant. ORC applications are expected to increase in landfill gas combustion plants in Turkey.

In this study, an industrial facility that manufactures MDF (Medium Density Fiberboard) and chipboard in wood sector in Turkey has been chosen as an example for an ORC application. Waste like wood chips, sawdust, bark, etc. is combusted in a thermal oil boiler to get thermal oil. A portion of thermal oil is used for the heat requirement of the process and the rest is used in the ORC unit to produce hot water and electricity. The produced electricity is sold to the grid by means of renewable energy incentives. Hot water is used for space heating in the winter and it also meets the needs of cooling for the process through a water-cooled absorption chiller in the summer.

In addition, there are various double-fluid (binary) ORC applications in geothermal plants can be found with high MW capacities in Turkey. Binary power plants use geothermal resources, which the temperature of these sources ranges between 150 °C and 180 °C. These power plants have been operating for 25 years and they are developing their technologies. The installed capacity of electricity generating plants from geothermal energy through ORC applications is 197 MWe in Turkey. The installed capacity of the geothermal based ORC plant that under construction is around 98 MWe. If the incentives for electricity production from geothermal sources continue to be given to the related sectors, it is strongly believed that the use of the ORC technology in geothermal based energy productions can rapidly spread and increase in the near future.

The production of electricity from waste heat through ORC applications is very new for Turkey. There are only a few pieces of plant that use ORC technologies to produce power from waste heat in the country. One of the main reasons of the failure to spread ORC applications is the lack of incentives for waste heat based power generation. Due to this reason, the period of redemption is around 5–8 years, which is very long time. Another important reason is ORC systems are not produced in Turkey. Thus, we presume that the production of electricity from waste heat through ORC applications will not be widespread much in the near future,

Table 14
The feasibility analysis of the ORC unit.

Facility Component	Calculation Method	Example	Price	Unit
The initial investment cost	Built-in capacity (kWh) × investment costs (€/kWh)	1000 × 1470	1,470,000	Euro (€)
Expenses				
The purchase of biomass	The consumption of biomass (kg/h) × biomass unit price (0.015 €/kg) × working hours (h/year)	2785 × 0.015 × 8000	334,200	€/year
Maintenance of ORC unit	Working hours/year × (2) €/h	8000 × 2	16,000	€/year
Insurance and taxes	Total investment costs × % 0.05	1,470,000 × 0.05	73,500	€/year
Labor force	Person × 12 month × 1,500 €	1 × 12 × 1500	18,000	€/year
The total annual costs	441,700			€
Incomes				
Electricity sales	Net electricity generation (kW) × working hours × unit price of electricity (0.07 €/ kWh)	995 × 8000 × 0.07	557,200	€/year
Heat Usage	The heat of condenser (kW) × working hours × 0.01 €/kWh	5450 × 8000 × 0.01	436,000	€/year
Annual total revenues	*993,200*			€
Annual benefit	(Incomes – Expenses)/year	(993,200–441,700)	551,500	€
Simple payback period (years)	Total investment cost/annual profit	1,470,000/551,500	**2.67**	**Year**

unless the productions of ORC systems are realized in Turkey and this kinds of productions should be supported by the state.

Law no: 6094 is a new renewable energy bill determining regulations and feed-in tariffs in the sector [58]. According to the law, producers of renewable energy who started operation between May 18, 2005 and December 31, 2015 will be guaranteed power purchase prices for a period of 10 years. Wind and hydro-electric power will be bought at USD 0.073 per kilowatt-hour (kWh), the price for geothermal energy is defined at USD 0.105 per kilowatt-hour (kWh), and biomass, including landfill gas, and solar energy will be purchased at USD 0.133 per kilowatt-hour (kWh). Renewable energy producers that got their operation licenses before December 31, 2015 are entitled to receive additional subsidy ranging from USD 0.004 to USD 0.035 per kWh for a period of five years if they use locally-produced equipment and technology for their plants [58].

Turkey's energy demand has increased rapidly over the last fifteen years and most probably will continue to increase in the near future, at this speed. About 50% of electricity generation is obtained from natural gas in Turkey and the vast majority of natural gas is imported from abroad. Due to the dependency on foreign sources for energy, Turkey needs to benefit from its own domestic renewable energy resources, such as biomass, geothermal and solar in a maximum level to generate power. Also, waste heat should be evaluated more for power production by using ORC systems in terms of providing high energy efficiency. With the realization of the manufacturing and using ORC technologies in Turkey, the reduction of energy import expenses will be assured. So, economic development of the country will be improved greatly.

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